

Rupesh Ghimire

AI/ML Engineer & Researcher | rupeshghimire17@gmail.com | +61 0415063505 | Mortdale, NSW

Portfolio | LinkedIn | Github | Medium

PROFESSIONAL SUMMARY

I am a registered Computer Engineer and AI Researcher who thrives on turning complex, messy data into intelligent systems that work. With nearly two-years of experience in the AI/ML industry, I've handled everything from fine-tuning LLMs using LoRA and DPO to building the MLOps pipelines that bring models to life. I've found that the best AI isn't just about the architecture; it's about the 'detective work'—using statistical rigor to uncover patterns that drive real business impact to provide real value to users and stakeholders. Whether I'm optimizing a transformer or mentoring a team, I focus on building technology that is as practical and reliable as it is innovative.

Education

Masters of Data Science July 2026 – 2028

Charles Darwin University, Haymarket, Australia, NSW

- **Coursework:**

Bachelors in Computer Engineering Oct 2019 – Apr 2024

Tribhuvan University, Institute of Engineering – Pashchimanchal Campus Percentage: 75.70%

Pokhara, Nepal

- **Coursework:** Computer Organization and Architecture, Theory of Computation, Artificial Intelligence, Digital Signal Analysis & Processing, Artificial Intelligence, Image Processing and Pattern Recognition, Software Engineering, Engineering Mathematics, Probability and Statistics

Micro Degree Jan 2023 – Jan 2024

Fusemachines - AI Fellowship

[Link](#)

REMOTE, Nepal

- **Coursework:** Machine Learning, Time Series Forecasting, Reinforcement Learning, Deep Learning, Computer Vision, Natural Language Processing, Language Models, Transformers

High School - Science: STEM May 2017 – May 2019

Nepal Education Board(NEB) - Nepal Mega College

GPA: 3.85/4.0

Kathmandu, Nepal

- **Coursework:** Physics, Chemistry, Mathematics, Computer Science

Competitions & Awards

1st Runner Up – Datathon, Delta Tech Fest 2023

1st Runner Up – Hult Prize OnCampus, IOE-Pokhara [Link](#) – 2020

Academic Achievements

Fusemachine AI Fellowship 2023-2024

Golden Jubilee Scholarship Scheme 2019-20 2019-2023

Merit Based Scholarship for Undergraduate Studies - Indian Embassy

Research & Technical Skills

Machine Learning & AI: PyTorch, HuggingFace, Transformer Architectures, LoRA, DPO, RAG, Knowledge Graphs

Research Focus: Federated Learning, Model Quantization, Custom Model Design,

Data & Experimentation: Data Preprocessing & Augmentation, STFT, LogMel, Evaluation Metrics, MLflow, Evidently AI, Literature Review

Backend & MLOps: Django, FastAPI, Docker, Nginx, Prefect, DVC, Server-based Pipeline Automation

Data Science & Tools: Pandas, NumPy, Scikit-learn, Matplotlib, LangChain, LlamaIndex, Vector Databases

Programming: Python, C, C++, SQL

Research & Experience

North Star Developer's Village (NSDEVIL) – AI/ML Engineer

Aug 2024 – Jun 2026

Kathmandu, Nepal (Remote - Republic of Korea)

- Designed and fine-tuned transformer-based image-to-text models using LoRA and PEFT methods for resource-constrained deployment.
- Built agentic RAG with memory using LangGraph, Langchain, pgvector, qdrant and ollama for internal documentation sharing with 98% uptime, reducing cross-function team collaboration delay by 40%.
- Built custom Transformer models in PyTorch for text generation and sequence-to-sequence tasks.
- Developed Federated Learning architecture for mobile and web clients, integrating model quantization for improved efficiency.
- Automated reproducible MLOps pipelines (MLflow, Prefect, DVC, Evidently AI) on company servers for experiment tracking and auditability.
- Containerized and deployed AI services (Django, FastAPI) as RESTful APIs with Docker and Nginx.
- Contributed to robotics R&D team through camera calibration and vision-based model development.

North Star Developer's Village (NSDEVIL) – AI Research Intern

May 2024 – Aug 2024

Kathmandu, Nepal

- Conducted literature reviews across NLP, audio processing, and multimodal learning to determine optimal architectures for company research.
- Performed prompt engineering for multimodal models to produce accurate, context-aware outputs.
- Designed and trained a custom audio classification model using STFT and LogMel features, achieving **90.01% accuracy** on a curated dataset.

Panacea Solution Pvt. Ltd. – Data Science Mentor

June 2023 - Feb 2024

Kathmandu, Nepal

- Provided Mentorship sessions for 9 months to 3 batches of students (nearly 50 students total) in a 3-months/60 hour session. Designed and tutored a well structured course using my AI/ML expertise to simplify complex concepts and help them build real projects in Python, Machine Learning, and introductory Deep Learning.

Selected Projects

Search Synthesizer

2024

RAG, LangChain, LlamaIndex, VectorDB, Web Scraping

- Built recursive LLM-based RAG pipeline with GoogleSearch API, content scraping, vector embeddings, and contextual media/sources.

Gantavya – Landmark Recognition

2024

YOLO, Transfer Learning, Django, React Native

- Created and augmented dataset for 15 landmark classes, achieving **97% accuracy** in detection/localization; deployed as mobile app with map navigation and JWT authentication.

Nepali Sentiment Analysis

2023

BERT, NLP, Flask API

- Preprocessed Nepali text (SpaCy/NLTK), fine-tuned NepBERTa to **76% accuracy** and deployed as real-time sentiment classification API.

Technical Reports

Gantavya – Landmark Recognition

May 2024

Undergraduate Major Project Report, Tribhuvan University

[Link](#)

Developed a landmark recognition system using YOLOv8, achieving **97% accuracy** on 15 classes; deployed as a mobile app with Django and React Native.

Volunteering & Outreach

ML Mentor – Innovative Computer Engineering Student’s Society	2024
Conducted workshops on Python, Scikit-learn, Pandas, and guided student ML projects.	Link
Django Mentor – Innovative Computer Engineering Student’s Society	2023
Conducted training workshop on Python, Django, Database, ORM and RESTful APIs.	Link

References

Bhuban Ghimire – Senior Backend Engineer (NSDEVIL)	Email
Ajay Yadav – Project Manager (NSDEVIL)	Email
Shushank Gyawali – UI/UX Lead Class Coordinator (Panacea Solution Pvt. Ltd.)	Email